

Seizure Relapse and Development of Drug Resistance Following Long-term Seizure Remission

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Objective: To quantify and identify predictive risk factors for seizure relapse and development of drug resistance in patients who achieved long-term (>1 year) antiepileptic drug-induced seizure remission.

Design: Prospective cohort study.

Setting: Epilepsy Center, Rambam Medical Center.

Patients: Two hundred fifty-six consecutive patients who entered long-term (>1 year) antiepileptic drug-induced seizure remission were followed up prospectively for 2 years or more.

Main Outcome Measures: Seizure relapse and development of drug-resistant epilepsy.

Results: Five years after entering seizure remission, 40.2% of patients experienced seizure relapse and 25.3% of patients developed drug-resistant epilepsy. The Kaplan-

Meier curves could be fitted by monoexponential functions, with a maximal seizure relapse rate of 43.6%, maximal drug-resistance rate of 27.4%, and half-decay constant of 21.5 months for both curves. Treatment history served as a significant independent prognostic risk factor for both seizure relapse and development of drug resistance. The preremission seizure frequency and duration of epilepsy were also identified as significant prognostic risk factors in the univariate analysis but failed to reach statistical significance in the multivariate analysis.

Conclusion: Seizure relapse commonly occurs in patients following long-term seizure remission. Treatment history and duration of epilepsy are predictive risk factors for both seizure relapse and development of drug resistance.

Arch Neurol. 2009;66(10):1233-1239

EPILEPSY MANIFESTS CLINICALLY as recurrent unprovoked seizures.^{1,2} Previous studies have shown that antiepileptic drug (AED) treatment can induce long-term seizure remission in up to 70% of epileptic patients.¹⁻¹⁴ The remaining 30% of epileptic patients have drug-resistant epilepsy and continue to experience seizures despite appropriate AED treatment.^{3-10,15-19} In contrast to the numerous studies that addressed the probability and prognostic risk factors for obtaining AED-induced seizure remission, few data exist concerning the long-term prognosis of epileptic patients who entered AED-induced seizure remission. Previous studies have reported that epileptic patients who entered long-term AED-induced seizure remission can experience seizure relapse and even develop drug-resistant epilepsy.^{14-16,18,20} However, the extent and time course of these phenomena are largely unknown. In addition, there are few data concerning the prognostic risk factors for these occurrences.

In this study, we prospectively followed up 256 consecutive patients who entered AED-induced long-term (>1 year) seizure remission to quantify the extent and time course of seizure relapse and development of drug-resistant epilepsy in these patients and define prognostic risk factors that can help identify the patients at risk for these occurrences.

METHODS

This study was performed on a cohort of 566 consecutive patients (≥12 years) who received newly administered AED treatments at the Epilepsy Clinic of the Rambam Medical Center between January 1999 and December 2005.¹⁹ Our clinic serves a diverse patient population, including patients with new-onset epilepsy, easy-to-treat epilepsy, and drug-resistant epilepsy.¹⁹ All patients included in our analysis were followed up prospectively for a minimal period of 2 years (range, 2-8.5 years; mean [SD], 45.2 [16.9] months) after the new AED treatment was administered. The mean [SD] age of the patients was 31.8 [15.8] years, with 11 patients younger than 16 years.

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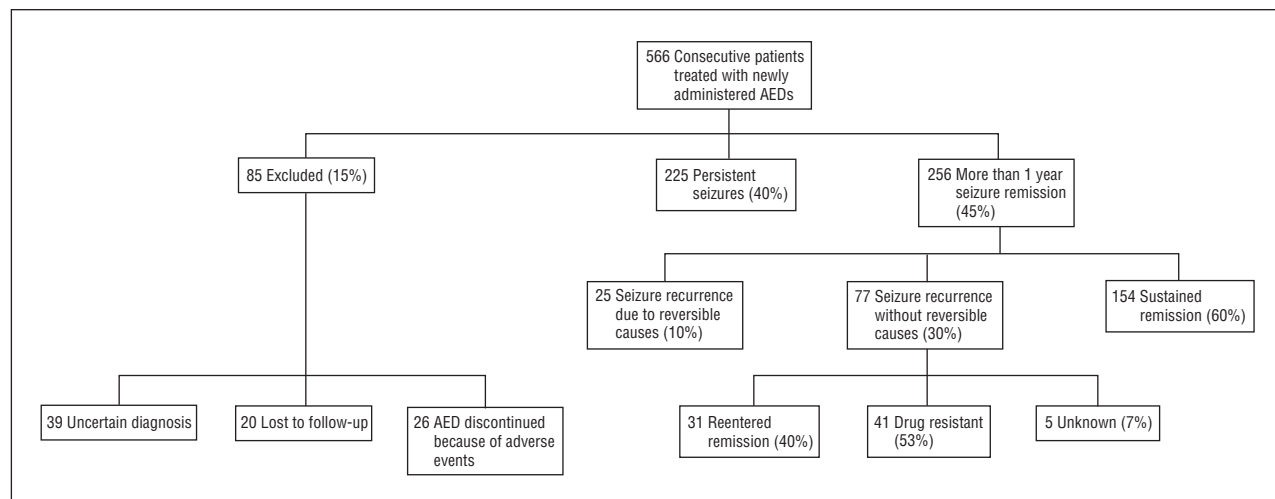


Figure 1. Flowchart of the patients in the study. AED indicates antiepileptic drug.

At the first and subsequent clinic visits, data were collected from every patient, family members, medical records, and letters from the primary treating physicians. In the first clinic visit, data were collected regarding the sex, age, age at onset of epilepsy, family history of epilepsy, risk factors of epilepsy, medical and neurological history, classification of present and past seizures, frequency of each seizure type, number of seizures in the past 3 months, classification of epilepsy and epileptic syndrome, electroencephalographic (EEG) and imaging data obtained prior to initiation of the newly administered AED treatment, present and past AED treatment, reasons for discontinuation of past AED treatment, and past seizure control. Classification of epileptic seizures, epilepsy type, and epileptic syndrome was performed according to International League Against Epilepsy Classification.^{21,22} On subsequent visits after initiation of newly administered AED treatments, we collected data concerning the number of seizures, adverse events, and compliance since the last clinic visit. In addition, regular blood testing was performed, including drug levels whenever possible (valproic acid, carbamazepine, and lamotrigine). All data were recorded in our in-house Oracle-based software specially modified for the Epilepsy Clinic.

Administration of AED treatments was done according to the clinical judgment of the treating epileptologists.¹⁹ We did not have a strict treatment algorithm for choosing the new AED treatment. However, in general, the first 2 AED treatments were prescribed as monotherapy. After 2 AEDs proved inefficient, we usually added a second AED to the existing AED. We tried to avoid simultaneous therapy with 3 drugs, but in some patients, triple therapy was unavoidable. Regarding the dose of AEDs, on initiation of a new AED treatment, the dose was usually increased to a medium dose range (typically 200 mg for lamotrigine and 100-200 mg for topiramate, 600-1000 mg for carbamazepine, 1200-1800 mg for gabapentin, and 600-1000 mg for valproic acid). If seizures recurred, the AED dose was further increased up to the maximally tolerated dose.

Patients attended regular clinic visits and, if needed, were contacted via telephone. Patients were asked to maintain seizure diaries. The number of seizures experienced since the last clinic visit was obtained from the seizure diaries and from the interviews held during the clinic visits.

In all cases of seizure relapse, we took special care to determine compliance of patients to AED treatment and identify other reversible external triggers for seizures. Diagnosis of non-compliance was based on self-reporting or AED blood levels (when AED blood levels after the seizure were 50% or lower than the previous levels).

In patients who experienced seizure recurrence, we first increased the dose of the administered AED. In cases where the AED dosage could not be further increased, a new AED was administered either as a replacement or add-on treatment.

Statistical analysis was performed using the χ^2 test, Fisher exact test, and logistic regression multivariate analysis. For multivariate analysis, a stepwise selection of the variables was used. The patient's sex, type of epilepsy, and EEG findings were entered as categorical dummy variables. The number of past ineffective AED treatments, duration of epilepsy, and number of seizures in the 3-month period prior to initiation of the new AED treatment were subcategorized and entered as dummy variables. All statistical analysis was performed using SPSS software (SPSS Inc, Chicago, Illinois). All graphs and curve fittings were performed using Igor software (WaveMetrics, Inc, Lake Oswego, Oregon).

RESULTS

A cohort of 566 consecutive patients received newly administered AED treatments between January 1999 and December 2005 in the Epilepsy Clinic of the Rambam Medical Center (**Figure 1**). Eighty-five patients were excluded from analysis because they were either lost to follow-up within 24 months of AED treatment initiation, discontinued all (1-3) new AED treatments prescribed because of adverse events within the first year of use, or had an uncertain diagnosis (possible nonepileptic spells). Of the remaining 481 patients, 256 patients (53%) entered long-term seizure remission (>1 year) and 225 patients (47%) continued to experience seizures (Figure 1). Of the 256 patients who entered long-term seizure remission after receiving the newly administered AED treatment, 172 (67%) received a single AED (monotherapy) and 85 patients (33%) received the newly administered AED as an add-on treatment to 1 (75 patients), 2 (6 patients), or 3 (4 patients) additional AEDs.

Sixty-four (25%) of these 256 patients had generalized epilepsy (62 with primary generalized epilepsy, 25 of whom had juvenile myoclonic epilepsy), 71 had cryptogenic partial epilepsy, and 91 had symptomatic partial epilepsy (23, cerebrovascular disease; 11, posttraumatic; 11, meningioma; 9, vascular malformations; 8,

hippocampal atrophy; 7, cortical dysplasia). The classification of the remaining 30 patients was undetermined. Eighty-one patients were newly treated (62 had ≥ 2 seizures and 19 patients had only 1 seizure prior to AED treatment initiation. The decision to initiate AED treatment after the first seizure was determined after consulting the patients and taking into account results of the EEG recordings). Prior to entering long-term seizure remission, 2 patients previously underwent unsuccessful epilepsy surgery, and none of the patients underwent vagal nerve stimulator implantation.

At the end of the follow-up period, 210 of 256 patients (82%) were seizure free and 41 patients (16%) developed drug-resistant epilepsy (Figure 1). One hundred fifty-four patients (60%) remained seizure free throughout the entire follow-up period, and 25 patients (10%) experienced seizure relapse due to external reversible causes and later regained seizure remission. In 77 patients (30%), seizures recurred without known reversible triggers. Forty-one of these 77 patients (53%) developed drug-resistant epilepsy and 31 patients (40%) reentered seizure remission either after dose adjustment (14 patients) or after changing the AED regimen (17 patients). In the remaining 5 patients (7%), the follow-up period was insufficient to determine whether patients could reenter seizure remission (Figure 1). Of the 41 patients who developed drug-resistant epilepsy, 13 had 1 additional AED treatment fail after seizure relapse, 23 patients had 2 AED treatments fail after seizure relapse, and the remaining 5 patients had 3 additional AED treatments fail after seizure relapse. All patients who developed drug resistance underwent surgical evaluation. Six patients were found to be good surgical candidates (2 underwent surgery, 2 declined surgery, and 2 are still in the process).

Figure 2 presents the Kaplan-Meier survival curves for seizure relapse and development of drug resistance as a function of the follow-up period. Five years after entering seizure remission, approximately 40% of patients experienced seizure relapse and 25% developed drug-resistant epilepsy. The curves for seizure relapse and development of drug resistance could be fitted by monoexponential functions, with a maximal seizure relapse rate of 43.6% and a maximal drug-resistance rate of 27.4%. Both curves had a half-decay constant of 21.5 months (Figure 2). Twenty-five additional patients experienced seizure relapse due to external reversible triggers, including discontinuation of AED treatment (7 patients), reduction of the AED dose (8 patients), noncompliance (7 patients), severe sleep deprivation (2 patients), and high fever (1 patient). These patients later regained seizure remission.

We next looked for predictive risk factors for seizure relapse and development of drug resistance after achieving long-term (>1 year) seizure remission. Using univariate analysis, we found 2 significant predictive risk factors for both seizure relapse and development of drug resistance: treatment history and duration of epilepsy (**Figure 3** and **Figure 4**) (**Table 1**). In addition, the preremission seizure frequency was identified as a significant predictive risk factor for development of drug resistance, but not for seizure relapse (Table 1).

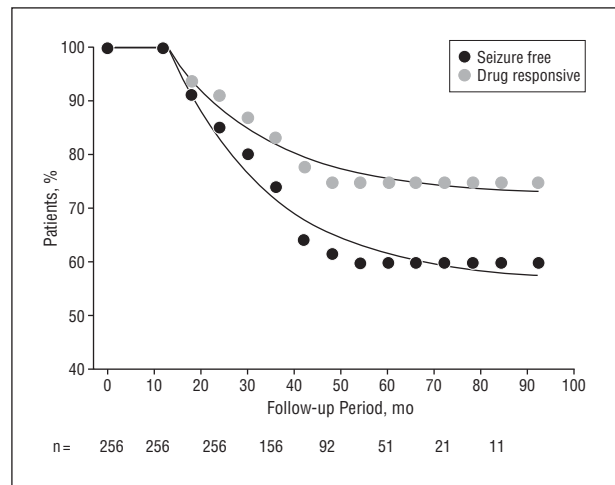


Figure 2. Seizure relapse and development of drug-resistant epilepsy in patients who entered long-term (>1 year) seizure remissions. The average percentages of patients who were seizure free and drug responsive are plotted as a function of time from the onset of the antiepileptic drug-induced seizure remission. The experimental points were fitted with monoexponential functions (lines), with minimal values of 56.4% for seizure relapses and 72.6% for development of drug resistance and a half-decay time constant of 21.5 months for both curves.

The probability for both seizure relapse and development of drug resistance was significantly higher in patients who had multiple ineffective AED treatments fail in the past. Patients who had 2 to 3 past ineffective AED treatments had a 2.1-fold (range, 1.1- to 3.2-fold) increase in seizure relapse and a 4.3-fold (range, 2.1- to 16.8-fold) increase in the probability to develop drug-resistant epilepsy compared with patients who had only zero to 1 past ineffective AED treatments (Figure 3) (Table 1). The rates of seizure relapse and development of drug resistance were even higher in patients who had 4 to 7 past ineffective AED treatments (Figure 3) (Table 1).

The duration of epilepsy at the time of remission also served as a significant prognostic risk factor for seizure relapse and development of drug-resistant epilepsy. Patients with preremission epilepsy durations of 5 years or more had a 1.5- to 2-fold increase in the seizure relapse rate and a 2.5- to 3-fold increase in the probability to develop drug resistance as compared with patients with a preremission epilepsy duration of less than 5 years (Figure 4) (Table 1).

The seizure frequency during the 3 months prior to seizure remission was a significant prognostic risk factor for development of drug resistance. Because the preremission seizure frequency was higher, the likelihood for seizure relapse and development of drug resistance increased (Table 1).

Other parameters we examined, including age, sex, type of epilepsy (partial vs generalized), etiology of epilepsy (idiopathic, cryptogenic, symptomatic), and EEG findings, did not significantly influence seizure relapse or development of drug resistance in our patients. Because of the small number of patients, we were unable to examine seizure relapse and development of drug resistance in specific epileptic syndromes.¹⁵

To further characterize independent prognostic risk factors for seizure relapse and development of drug-

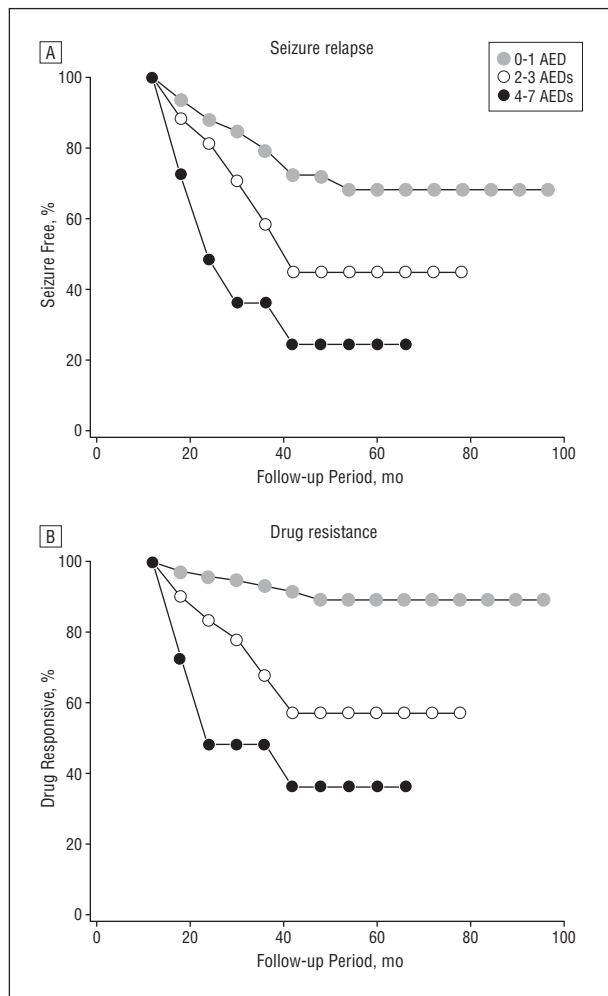


Figure 3. The average percentages of patients who were seizure free (A) and drug responsive (B) are plotted as a function of time from the onset of antiepileptic drug (AED)-induced seizure remission in patients who had 0 to 1, 2 to 3, and 4 to 7 past ineffective AED treatments. Probability for seizure relapse and development of drug-resistant epilepsy increase as a function of the number of past AED treatments that failed due to inefficacy. Drug-responsive patients are defined as patients without drug-resistant epilepsy.

resistant epilepsy in patients who entered long-term (>1 year) seizure remission, we performed logistic regression multivariate analysis (**Table 2**). The multivariate analysis confirmed that the treatment history served as a significant independent predictive risk factor for both seizure relapse and development of drug-resistant epilepsy (Table 2). In contrast to the univariate analysis, multivariate analysis failed to reach statistical significance for the preremission seizure frequency and epilepsy duration. In addition, it confirmed that the age, sex, type of epilepsy (partial vs generalized), etiology of epilepsy (idiopathic, cryptogenic, symptomatic), and EEG findings did not serve as significantly independent risk factors for either seizure relapse or development of drug resistance in patients who entered AED-induced long-term (>1 year) seizure remission (Table 2).

COMMENT

In this study, we characterized the long-term prognosis of epileptic patients who entered long-term (>1

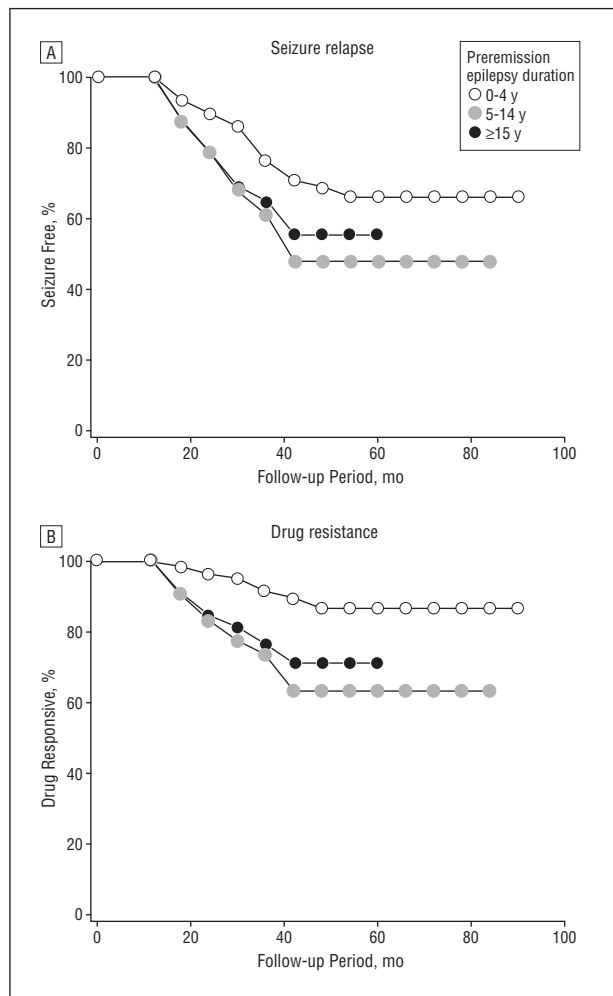


Figure 4. The effect of the preremission epilepsy duration on seizure relapse and development of drug-resistant epilepsy in patients who entered long-term (>1 year) seizure remission. The average percentages of patients who were seizure free (A) and drug responsive (B) are plotted as a function of time from the onset of the AED-induced seizure remission in patients with preremission epilepsy duration of 0 to 4 years, 5 to 14 years, and 15 years or more. The rates of seizure relapse and development of drug-resistant epilepsy increase in patients with preremission epilepsy durations of 5 years or more.

year) seizure remission. The main findings of this study are:

1. Five years after obtaining drug-induced seizure remission, 40.2% of patients experienced seizure relapse and 25.3% developed drug-resistant epilepsy. The Kaplan-Meier survival curves describing seizure relapse development of drug resistance could be fitted by monoexponential functions, with a maximal relapse rate of 43.6%, maximal drug-resistance conversion rate of 27.4%, and half-decay time constant of 21.5 months for both curves. Hence, with time, almost half of the patients who entered long-term seizure remission are expected to experience seizure relapse, and more than a quarter of them are expected to develop drug-resistant epilepsy. In the majority of patients, seizure relapse and development of drug-resistant epilepsy is expected to occur within 4 to 5 years after entering seizure remission. However, the number of patients followed up for longer periods was small (Figure 2), and hence, it is possible that there is a

Table 1. Effect of Various Factors on Seizure Relapse and Development of Drug-Resistant Epilepsy: Univariate Analysis

Variable	No. of Patients	Seizure Recurrence, %	RR (95% CI)	P Value	No. of Patients	Drug Resistance, %	RR (95% CI)	P Value
Sex								
M	115	30	1 [Reference]	.92	115	18	1 [Reference]	.38
F	141	31	1.0 (0.6-1.5)		141	14	0.8 (0.4-1.7)	
No. of past ineffective AED treatments								
0-1	185	20	1 [Reference]	.001	185	7	1 [Reference]	<.001
2-3	57	42	2.1 (1.1-3.2)		57	30	4.3 (2.1-16.8)	
4-7	14	64	3.2 (1.2-6.5)		14	57	8.1 (2.2-19.6)	
Type of epilepsy								
Generalized	64	25	1 [Reference]	.36	64	11	1 [Reference]	.14
Partial	162	32	1.28 (0.8-1.8)		162	19	1.7 (0.5-3.1)	
Etiology of epilepsy								
Idiopathic	68	19	1 [Reference]	.12	68	7	1 [Reference]	.91
Cryptogenic	81	30	1.58 (0.8-4.2)		81	21	3 (1.4-4.9)	
Symptomatic	105	33	1.73 (1.2-2.5)		105	19	2.7 (1.4-4.3)	
Duration of epilepsy, y								
0-4	144	21	1 [Reference]	.03	144	9	1 [Reference]	<.001
5-14	66	30	1.4 (0.8-2.2)		66	27	g3 (1.4-6)	
≥15	46	39	1.9 (0.7-2.8)		46	24	2.6 (1-4.8)	
No. of seizures in past 3 mo								
0-1	108	23	1 [Reference]	.23	108	9	1 [Reference]	.006
2-5	98	35	1.5 (0.9-2.1)		98	17	1.9 (1-3.3)	
6-10	25	36	1.6 (0.5-2.8)		25	28	3.1 (0.8-6.2)	
>10	25	36	1.6 (0.5-2.8)		25	32	3.6 (1-6.7)	
EEG findings								
Normal	55	24	1 [Reference]	.29	55	16	1 [Reference]	>.99
Epileptiform	124	31	1.3 (0.7-1.9)		124	16	0.97 (0.92-1.03)	

Abbreviations: AED, antiepileptic drug; CI, confidence interval; EEG, electroencephalographic; RR, relative risk.

Table 2. Effect of Various Factors on Seizure Relapse and Development of Drug-Resistant Epilepsy: Multivariate Analysis

Variable	Seizure Relapse		Drug Resistance	
	OR (95% CI)	P Value	OR (95% CI)	P Value
Sex				
M	1 [Reference]	.96	1 [Reference]	.76
F	0.97 (0.4-2.3)		0.8 (0.3-4.7)	
No. of past ineffective AED treatments				
0-1	1 [Reference]	.01	1 [Reference]	<.01
2-3	4.1 (1.4-12)		15.1 (3.1-74.5)	
4-7	5.4 (0.6-51.5)		9.7 (0.56-167.5)	
Type of epilepsy				
Generalized	1 [Reference]	.21	1 [Reference]	.86
Partial	3.2 (0.5-20.4)		1.24 (0.2-9.4)	
Etiology of epilepsy				
Idiopathic	1 [Reference]	.42	1 [Reference]	.21
Cryptogenic	1.1 (0.4-2.5)		2.9 (0.5-17)	
Symptomatic	1.4 (0.6-3.2)		2.5 (0.4-13.6)	
Duration of epilepsy, y				
0-4	1 [Reference]	.86	1 [Reference]	.94
5-14	1.3 (0.6-3)		1.6 (0.4-8.2)	
≥15	1.3 (0.5-3.4)		1.1 (0.2-7.7)	
No. of seizures in past 3 mo				
0-1	1 [Reference]	.96	1 [Reference]	.23
2-5	1.3 (0.5-3.5)		1.1 (0.3-4.6)	
6-10	1.0 (0.3-5.2)		5.5 (0.8-35.4)	
>10	0.9 (0.2-4.3)		3.1 (0.5-17.6)	
EEG findings				
Normal	1 [Reference]	.12	1 [Reference]	.30
Epileptiform	1.9 (0.6-5.5)		1.9 (0.4-9.4)	

Abbreviations: AED, antiepileptic drug; CI, confidence interval; EEG, electroencephalographic; OR, odds ratio.

second, slower process of seizure relapse and development of drug resistance that cannot be detected in our cohort. An additional 10% of patients experienced seizures due to reversible external triggers such as non-compliance, dose reduction, or discontinuation of the AED treatment.

2. The treatment history (number of past ineffective AED treatments) served as an independent predictive risk factor for both seizure relapse and development of drug-resistant epilepsy. In addition, the preremission seizure frequency and duration of epilepsy were found to be significant predictive risk factors in the univariate statistical analysis but failed to reach statistical significance on the multivariate analysis.

Antiepileptic drug-induced seizure remissions have been extensively studied by previous authors.^{4-7,10-16,18-20,23-25} In contrast, only few previous studies addressed the questions of seizure relapse and development of drug-resistant epilepsy in patients who entered seizure remission, especially in adults.^{14-16,18,20}

Mohanraj and Brodie¹⁶ reported that 20% of patients with newly diagnosed epilepsy who entered long-term seizure remission eventually relapsed and 8% eventually developed drug resistance. However, this study did not analyze the data with Kaplan-Meier survival curves and failed to identify prognostic risk factors for these occurrences. The seizure relapse rates reported by Mohanraj and Brodie were significantly lower than those we found, probably because the vast majority of patients in the Glasgow cohort had only zero to 1 past ineffective AED treatments.¹⁶

Choi et al¹⁸ reported that 5 of 20 patients who previously had drug-resistant epilepsy and entered seizure remission eventually relapsed. The small number of patients who entered seizure remission limited the conclusions of this study.

Three additional studies have been performed on pediatric cohorts. In a long-term population-based study from Finland,¹⁴ 47% of patients who entered seizure remission relapsed and 17% eventually developed drug resistance. In 2 large studies, Berg et al^{15,20} reported that approximately 30% of patients with drug-resistant epilepsy experienced delayed development of drug resistance and that approximately 20% to 25% of these patients entered transient long-term seizure remission prior to development of drug resistance. Several points differentiate our study from these 2 large-scale studies.^{15,20} First, these studies were performed on pediatric cohorts, while our cohort consisted mostly of adults. Second, these studies did not investigate the outcome of patients who entered long-term remission but rather concentrated on the time course for development of drug-resistant epilepsy.²⁰ Third, these 2 studies only looked at drug resistance but did not investigate seizure relapse that was later controlled.

There are several possible underlying theoretical mechanisms for delayed seizure relapse and development of drug resistance in patients who entered seizure remission. First is the development of drug tolerance to AEDs.^{26,27} Drug tolerance can result from various pharmacodynamic mechanisms that reduce the antiepileptic efficacy of AEDs, such as changes in the sensitivity of volt-

age-gated channels or other target molecules to AEDs²⁶⁻²⁸ or upregulation of drug transporters that enhance extraction of AEDs from neurons.^{29,30} The second mechanism that can explain delayed seizure relapse and development of drug resistance is progression of the underlying epileptogenesis.^{26,27}

The most powerful predictive risk factor for both seizure relapse and development of drug-resistant epilepsy was the number of past ineffective AED treatments. In a previous study, we showed that the number of past ineffective AED treatments was also the most powerful predictive risk factor for obtaining the initial seizure remission in this cohort.¹⁹

Other predictive factors for obtaining the initial seizure remission, including the preremission seizure frequency and duration of epilepsy, were also found to predict seizure relapse and development of drug resistance on univariate analysis but failed to reach statistical significance in the multivariate analysis. Taken together, these findings suggest that seizure relapse and development of drug resistance are dependent on the initial severity of epilepsy.

The difference between the univariate and multivariate analyses may be because patients with higher seizure frequency or longer epilepsy duration had taken more AEDs in the past.

Even though patients were followed up prospectively after administration of new AED treatments, some data concerning the history of their epilepsy were obtained retrospectively. This included data regarding treatment history, duration of epilepsy, and number of past seizures. It will, however, be virtually impossible to perform a similar study in a cohort of new-onset epilepsy because of the long follow-up required and the fact that the majority of patients will respond well to the initial AED. Indeed, in a large cohort of newly diagnosed patients, the number of patients who developed drug-resistant epilepsy was small, and no patients had prolonged epilepsy or multiple past AED treatments that failed.^{7,16}

In conclusion, in this study, we examined a question that is frequently raised in the clinical practice of every epileptologist: What is the prognosis of patients who entered long-term seizure remission? We found that seizure relapse was common. Slightly less than half the patients who entered long-term (>1 year) AED-induced seizure remission relapsed and more than a quarter of them eventually developed drug-resistant epilepsy. In the vast majority of patients, seizure relapse and drug resistance occurred within the first 4 to 5 years after obtaining seizure remission. We found that the treatment history served as an independent prognostic risk factor for development of drug-resistant epilepsy. The pretreatment seizure frequency and duration of epilepsy were also identified as significant prognostic risk factors on univariate analysis, but did not reach statistical significance on the multivariate analysis. The data presented herein will assist in determining the prognosis of long-term AED treatment and the pros and cons for referral to epilepsy surgery.

Accepted for Publication: April 15, 2009.

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Financial Disclosure: None reported.

REFERENCES

- Browne TR, Holmes GL. Epilepsy. *N Engl J Med*. 2001;344(15):1145-1151.
- Duncan JS, Sander JW, Sisodiya SM, Walker MC. Adult epilepsy. *Lancet*. 2006;367(9516):1087-1100.
- Brodie MJ, Dichter MA. Antiepileptic drugs. *N Engl J Med*. 1996;334(3):168-175.
- Annegers JF, Hauser WA, Elveback LR. Remission of seizures and relapse in patients with epilepsy. *Epilepsia*. 1979;20(6):729-737.
- Cockerell OC, Johnson AL, Sander JW, Hart YM, Shorvon SD. Remission of epilepsy: results from the National General Practice Study of Epilepsy. *Lancet*. 1995;346(8968):140-144.
- Mattson RH, Cramer JA, Collins JF. Prognosis for total control of complex partial and secondarily generalized tonic clonic seizures: Department of Veterans Affairs Epilepsy Cooperative Studies No. 118 and No. 264 Group. *Neurology*. 1996;47(1):68-76.
- Kwan P, Brodie MJ. Early identification of refractory epilepsy. *N Engl J Med*. 2000;342(5):314-319.
- Elger CE. Pharmacoresistance: modern concept and basic data derived from human brain tissue. *Epilepsia*. 2003;44(suppl 5):9-15.
- Schmidt D, Loscher W. Drug resistance in epilepsy: putative neurobiological and clinical mechanisms. *Epilepsia*. 2005;46(6):858-877.
- Callaghan BC, Anand K, Hesdorffer D, Hauser WA, French JA. Likelihood of seizure remission in an adult population with refractory epilepsy. *Ann Neurol*. 2007;62(4):382-389.
- MacDonald BK, Johnson AL, Goodridge DM, Cockerell OC, Sander JW, Shorvon SD. Factors predicting prognosis of epilepsy after presentation with seizures. *Ann Neurol*. 2000;48(6):833-841.
- Dlugos DJ, Sammel MD, Strom BL, Farrar JT. Response to first drug trial predicts outcome in childhood temporal lobe epilepsy. *Neurology*. 2001;57(12):2259-2264.
- Stephen LJ, Kwan P, Brodie MJ. Does the cause of localization-related epilepsy influence the response to antiepileptic drug treatment? *Epilepsia*. 2001;42(3):357-362.
- Sillanpää M, Schmidt D. Seizure clustering during drug treatment affects seizure outcome and mortality of childhood-onset epilepsy. *Brain*. 2008;131(pt 4):938-944.
- Berg AT, Vickrey BG, Testa FM, et al. How long does it take for epilepsy to become intractable? a prospective investigation. *Ann Neurol*. 2006;60(1):73-79.
- Mohanraj R, Brodie MJ. Diagnosing refractory epilepsy: response to sequential treatment schedules. *Eur J Neurol*. 2006;13(3):277-282.
- French JA. Refractory epilepsy: clinical overview. *Epilepsia*. 2007;48(suppl 1):3-7.
- Choi H, Heiman G, Pandis D, et al. Seizure remission and relapse in adults with intractable epilepsy: a cohort study. *Epilepsia*. 2008;49(8):1440-1445.
- Schiller Y, Najjar Y. Quantifying the response to antiepileptic drugs: effect of past treatment history. *Neurology*. 2008;70(1):54-65.
- Berg AT, Langfitt J, Shinnar S, et al. How long does it take for partial epilepsy to become intractable? *Neurology*. 2003;60(2):186-190.
- Commission on Classification and Terminology of the International League Against Epilepsy. Proposal for revised classification of epilepsies and epileptic syndromes. *Epilepsia*. 1989;30(4):389-399.
- Commission on Classification and Terminology of the International League Against Epilepsy. Proposal for revised clinical and electroencephalographic classification of epileptic seizures. *Epilepsia*. 1981;22(4):489-501.
- Camfield C, Camfield P, Gordon K, Dooley J. Does the number of seizures before treatment influence ease of control or remission of childhood epilepsy? not if the number is 10 or less. *Neurology*. 1996;46(1):41-44.
- Semah F, Picot MC, Adam C, et al. Is the underlying cause of epilepsy a major prognostic factor for recurrence? *Neurology*. 1998;51(5):1256-1262.
- Kim WJ, Park SC, Lee SJ, et al. The prognosis for control of seizures with medications in patients with MRI evidence for mesial temporal sclerosis. *Epilepsia*. 1999;40(3):290-293.
- Avanzini G. Is tolerance to antiepileptic drugs clinically relevant? *Epilepsia*. 2006;47(8):1285-1287.
- Löscher W, Schmidt D. New horizons in the development of antiepileptic drugs: innovative strategies. *Epilepsy Res*. 2006;69(3):183-272.
- Remy S, Gabriel S, Urban BW, et al. A novel mechanism underlying drug-resistance in chronic epilepsy. *Ann Neurol*. 2003;53(4):469-479.
- Lazarowski A, Czornyj L, Lubienieki F, Girardi E, Vazquez S, D'Giano C. ABC transporters during epilepsy and mechanisms underlying multidrug resistance in refractory epilepsy. *Epilepsia*. 2007;48(suppl 5):140-149.
- Löscher W. Drug transporters in the epileptic brain. *Epilepsia*. 2007;48(suppl 1):8-13.

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Special Theme Issue: Cancer and the Nervous System

Archives will publish a special theme issue in March 2010 on Cancer and the Nervous System. We invite submission of papers as Neurological Reviews, Clinical Trials, Original Communications, Case Reports, Images in Neurology, and Research Letters. Papers submitted by October 1, 2009, will have the best opportunity to be considered for this theme issue.